GABBER'S THEOREM IN H^2 FORM, INTERNALLY

Abstract. In synthetic algebraic geometry with the root topology (a refinement of the étale topology, Section 1), we prove: for a finitely presented A and a family G of small $R^\times$-gerbes over $\operatorname{Spec} A$ which are stacks, if $k \cdot \mathrm{cl}(G) = 0$ in the group H^2 for some $k \geq 1$, then there merely is an Azumaya algebra E over $\operatorname{Spec} A$ with $\delta(E) = \mathrm{cl}(G)$ in H^2. The gerbe form, an Azumaya algebra E whose splittings are fully faithfully compared with G, was obtained before. Here we add the splitting gerbe $\mathrm{Split}(E)$, show that it is a stack and a gerbe, define H^2 as a set truncation of maps into stack gerbes, and show that the comparison $G \to \mathrm{Split}(E)$ is an equivalence of banded stacks, so that the classes are equal by univalence. The group law on H^2 is the contracted product, built from families of rank-one modules of a given weight; $k \cdot \mathrm{cl}(G) = 0$ holds exactly when the band change k_*G has a global object. The sheafification $\Diamond$ is constructed (by towers of root adjunctions), not assumed, and the splitting gerbe is made small by replacement, so that the gerbes of the hypothesis and those of H^2 live in one universe. The whole argument is checked in Unit.

1. Setting

We work in univalent type theory with function extensionality, propositional truncation $\|-\|$ and set quotients (both obtained from replacement), over a local ring R satisfying the axioms of synthetic algebraic geometry. Universes are $\mathcal{U}_0 \subset \mathcal{U}_1$; R and modules live in $\mathcal{U}_0$.

Topology. For a monic polynomial f of positive degree let $G_f = \|\{x \mid f(x) = 0\}\|$, a proposition. (Degree 0 is excluded: $f = 1$ would give G_f empty and make every sheaf contractible.) We call the topology generated by the G_f the *root topology*. It is the one used in [4], Remark 1.1.2: the étale topology is generated by the G_f with f unramifiable in the sense of Wraith, and this condition is never used, so all monic f of positive degree are allowed. The root topology is therefore finer than the étale topology; it contains, e.g., the root of $T^p - a$ in characteristic p, which is not an étale cover. Everything below, including the definition of H^2, is stated and checked for the root topology.

A type X is a *sheaf* if for every f the constant map $X \to (G_f \to X)$ is an equivalence. Sheaves are closed under Π, Σ, identity types and retracts, and R is a sheaf. For a type P we write $\Diamond P$ for the sheafification of $\|P\|$ ("P holds locally"), constructed as follows.

A *tower* over a type Y is either empty, with points Y, or a monic polynomial c_y of positive degree depending on $y : Y$ followed by a tower over $\sum_{y:Y} \{x \mid c_y(x) = 0\}$; its points are those of the last stage, each lying over a point of Y. For $P \in \mathcal{U}_0$ let

$$\Diamond P = \|\text{a tower over 1 all of whose points give } \|P\|\|.$$

Proposition 1.1 (the closure)**.** *$\Diamond P$ is a proposition and a sheaf, $P \to \Diamond P$, and $\Diamond P \to Q$ for every proposition Q (in any universe) which is a sheaf and receives P. The sheaves in $\mathcal{U}_0$ form a sheaf.*

Proof. $P \to \Diamond P$ by the empty tower. *Elimination*: by induction on the tower, a map from its points to Q gives a map from its base to Q; at a stage c, for each y a map from the roots of c_y to Q factors through G_{c_y}, and Q is a sheaf. *Sheaf*: given $G_f \to \Diamond P$, the points of $\operatorname{Spec} B$, $B = R[X]/(f)$, are the roots of f, so at each point of $\operatorname{Spec} B$ there is merely a tower; by Z-choice and the flattening of towers (locally constant shapes, clearing of denominators,

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products) there is merely one tower over $\operatorname{Spec} B$ with the same property, and prefixed by the stage f it is a tower over 1. *Sheaves form a sheaf*: a family $X : G_f \to \mathrm{Sheaf}$ glues to $\prod_{p:G_f} X_p$, a sheaf, and since G_f is a proposition evaluation at p is an equivalence $\prod_p X_p \simeq X_p$, so by univalence the glued sheaf restricts to X; a constant family X glues to $G_f \to X$, equal to X since X is a sheaf. □

Two facts are used repeatedly:

(1) a sheaf proposition that holds locally holds;
(2) if P is a proposition and $G_f \to P$ with P a sheaf, then P.

Modules. A *basis* of rank N of a module M is a family $(e_i, c_i)_{i<N}$ of vectors and linear forms with $c_i(e_j) = \delta_{ij}$ and $\sum_i c_i(x)e_i = x$. Equivalently, a family w such that $a \mapsto \sum a_i w_i$, $R^N \to M$, is an equivalence. We use

Lemma 1.2. *If M merely has a basis, the carrier of M is a sheaf.*

Proof. With a basis, M is a retract of R^N ($x \mapsto (c_i(x))$, $a \mapsto \sum a_i e_i$), and R^N is a sheaf. For a set, being a sheaf holds as soon as it merely holds. □

Theorem 1.3 (descent of freeness)**.** *A module whose carrier is a sheaf and which locally has a basis of rank N merely has a basis of rank N.*

This is Theorem D of [4] (1.3.5), for the root topology, proved earlier; we use it as a black box.

2. Gerbes, stacks and H^2

Definition 2.1. A *band* at a point x of a 1-type X is a group isomorphism $\beta_x : (x = x) \to R^\times$ (a map with contractible fibres turning composition of loops into products). A *stack gerbe* is a type $X \in \mathcal{U}_0$ with a band at every point, such that

- X is a 1-type, $\Diamond X$, and $\Diamond(a = b)$ for all a, b;
- each identity type $a = b$ is a sheaf;
- X itself is a sheaf (*descent for objects*).

A *gerbe* is the same without the last condition; a gerbe whose carrier is a sheaf is a stack gerbe.

Because the paths of a gerbe are sheaves and locally inhabited, they form an $R^\times$-torsor which is a sheaf, hence (Hilbert 90) any two objects are merely equal.

Proposition 2.2 (identity of stack gerbes)**.** *Let G, G' be stack gerbes and $e : G \simeq G'$ an equivalence of carriers preserving the bands: $\beta'_{e(x)}(\mathsf{ap}_e(\ell)) = \beta_x(\ell)$. Then $G = G'$.*

Proof. By induction on equivalences (univalence in $\mathcal{U}_0$) we may take e to be the identity. Then the bands agree pointwise ($\mathsf{ap}_{\mathrm{id}}\ell = \ell$), and the remaining data are propositions: being a 1-type, $\Diamond X$, $\Diamond(a = b)$, being a sheaf, and, for a given map $(x = x) \to R^\times$, having contractible fibres and being multiplicative. □

Definition 2.3. For a type X (here $X = \operatorname{Spec} A$), $H^2(X) = \|X \to \mathrm{StackGerbe}\|_0 \in \mathcal{U}_1$, the quotient of $X \to \mathrm{StackGerbe}$ by mere equality. For $G : X \to \mathrm{StackGerbe}$ we write $\mathrm{cl}(G)$ for its class.

By Proposition 2.2 and function extensionality, $\mathrm{cl}(G) = \mathrm{cl}(G')$ as soon as there is, at each x, a band-preserving equivalence $G_x \simeq G'_x$. By Theorem 5.9, StackGerbe is the sheafification of $K(R^\times, 2)$, so $H^2(X) = \|X \to \bigcirc K(R^\times, 2)\|_0$ is the second cohomology in the sense of homotopy type theory.

3. The splitting gerbe

Fix $N = n + 1 \geq 1$ and an R-algebra E.

Definition 3.1. A *splitting* of E is a module M which merely has a basis of rank N, with an algebra isomorphism $\sigma : E \cong \operatorname{End} M$. An *arrow* $(M, \sigma) \to (M', \sigma')$ is a linear bijection $a : M \to M'$ with $a(\sigma(s)x) = \sigma'(s)(ax)$ for all $s \in E$. Arrows are determined by their underlying map. $\operatorname{Split}(E) \in \mathcal{U}_1$ is the type of splittings (its objects are modules).

Proposition 3.2. *The map* $(o = o') \to \operatorname{Arr}(o, o')$ *sending* refl *to the identity is an equivalence. Hence* $\operatorname{Split}(E)$ *is a* 1*-type.*

Proof. It suffices that $\sum_{o'} \operatorname{Arr}(o, o')$ is contractible. An element is a module M' with a linear equivalence $e : M \simeq M'$, a mere basis and a splitting σ' of M' intertwined by e. By the structure identity principle for modules we may take $e = \mathrm{id}$. Then $\sigma(s)x = \sigma'(s)x$ for all s, x, so $\sigma = \sigma'$; the bases agree (a proposition), and the arrow is the identity. □

Proposition 3.3 (essentially small)**.** *Let* $D = \{\sigma : E \cong \operatorname{End}(R^N)\} \in \mathcal{U}_0$ *and* $f : D \to \operatorname{Split}(E)$*,* $f(\sigma) = (R^N, \sigma)$*. The image* $\operatorname{Split}_0(E) \in \mathcal{U}_0$ *of* f *given by replacement comes with an embedding* $j : \operatorname{Split}_0(E) \to \operatorname{Split}(E)$*, and* j *is an equivalence.*

Proof. Replacement applies since $\operatorname{Split}(E)$ is locally small: $o = o'$ is equivalent to $\operatorname{Arr}(o, o') \in \mathcal{U}_0$ (Proposition 3.2). The fibres of the embedding j are propositions; they are inhabited: given $o = (M, \sigma)$ we may choose a basis of M, i.e. $q : M = R^N$, and $(R^N, q_*\sigma) = (M, \sigma)$ by induction on q; so o is $f(q_*\sigma)$. □

Lemma 3.4 (centre)**.** *Let* M *have a basis of rank* $N \geq 1$ *and let* γ *be a linear bijection of* M *commuting with every endomorphism. Then* $\gamma = \mu$ *for a unique unit* μ*.*

Proof. Let $f_i = c_0(-)e_i$. Then $\gamma(e_i) = \gamma(f_i e_0) = f_i(\gamma e_0) = \mu e_i$ with $\mu = c_0(\gamma e_0)$, so $\gamma = \mu$ by linearity and $\sum c_i(x)e_i = x$. If $\gamma y = e_0$ then $1 = c_0(\mu y) = \mu c_0(y)$. Uniqueness: compare at e_0. □

Proposition 3.5 (band)**.** *For a splitting* o*, the scalars give an equivalence* $R^\times \simeq \operatorname{Arr}(o, o)$*, and the composite* $\beta_o : (o = o) \simeq \operatorname{Arr}(o, o) \simeq R^\times$ *is a band.*

Proof. An arrow a of o commutes with every endomorphism f of M, since $f = \sigma(s)$ for $s = \sigma^{-1}(f)$; apply Lemma 3.4 (the goal is a proposition, so we may use a basis). Multiplicativity: the arrow of a composite path is the composite of the arrows (by path induction), and the composite of the scalars u and v is uv. □

Theorem 3.6. *If the carrier of* E *is a sheaf and locally* $E \cong M_N(R)$*, then* $\operatorname{Split}_0(E)$ *is a stack gerbe.*

Proof. Paths are sheaves. By Lemma 1.2 the carriers of M, M' are sheaves, so $\operatorname{Arr}(o, o')$ is a sheaf: linear maps $M \to M'$ form a sheaf (Π and identity types), being a bijection is a retract of the data $(g, h, fg = 1, hf = 1)$, a sheaf, and intertwining is a Π of identities. By Proposition 3.2, $o = o'$ is a sheaf.

Locally inhabited. R^N with its standard basis gives $\operatorname{End}(R^N) \cong M_N(R)$; where $E \cong M_N(R)$ this is a splitting of E, an element of D.

Locally connected. Two splittings (M, σ), (M', σ') are merely isomorphic: choose bases, so that both give automorphisms of $M_N(R)$; by Skolem–Noether over the local ring R, $\sigma'\sigma^{-1}$ is merely a conjugation, which yields an arrow.

Band. Proposition 3.5.

Descent for objects. Let $f : G_g \to \mathrm{Split}(E)$ (G_g a generator, a proposition). Put

$$M = \prod_{p:G_g} M_{f(p)}, \qquad \sigma(s)(m)(p) = \sigma_{f(p)}(s)(m(p)).$$

This is the key construction: since G_g is a proposition, for each p evaluation at p is a linear bijection $M \cong M_{f(p)}$ carrying σ to $\sigma_{f(p)}$. Hence: M has a basis under the hypothesis $p : G_g$; its carrier is a sheaf (a product of sheaves); so by Theorem 1.3 it merely has a basis. Likewise σ is an equivalence under p, and being an equivalence between sheaves is a sheaf proposition, so σ is an equivalence. The glued splitting $o_f = (M, \sigma)$ restricts to f: evaluation at p is an arrow $o_f \to f(p)$, hence a path (Proposition 3.2). Conversely, for a constant family $f = \lambda p.\, o$, the constant map $M_o \to \prod_p M_o$ is a bijection because M_o is a sheaf, and it is an arrow $o \to o_f$. So $o \mapsto \lambda p.\, o$ is an equivalence.

The small model. All of this passes to $\mathrm{Split}_0(E)$ along the equivalence j (Proposition 3.3): its identity types are retracts of the arrows, it is a sheaf, and $\beta_a(\ell) = \beta_{ja}(\mathsf{ap}_j \ell)$ is a band since ap_j is an equivalence preserving composition. □

4. Gabber's theorem

Twisted modules. For a gerbe G and a homomorphism $\chi : R^\times \to R^\times$, a family $V : G \to \mathrm{Mod}$ has *weight* χ if each loop ℓ at g acts on V_g (by transport) as multiplication by $\chi(\beta_g \ell)$. The band change $\chi_* G$ is the type of weight-χ families of modules whose fibres merely have a basis of rank one; it is a stack gerbe banded by $R^\times$ (Proposition 5.1). Here we use its objects: a weight-k family of rank-one modules on G. By Section 5, $k_* G_x$ has an object for all x, merely and uniformly in x, exactly when $k \cdot \mathrm{cl}(G) = 0$.

Theorem 4.1 (gerbe form, proved before)**.** *Let A be finitely presented and G a family of gerbes over* $\mathrm{Spec}\, A$ *with an element of $\prod_{x:\mathrm{Spec}\, A} k_* G_x$. Then merely there are $N \geq 2$ and a weight-1 family V_x on G_x whose fibres merely have bases of rank N, such that the descended endomorphism algebra $E_x = \mathrm{End}(V_x)^{G_x}$ (invariant sections) is Azumaya (a sheaf algebra, locally isomorphic to $M_N(R)$), evaluation at g is an algebra isomorphism $\mathrm{ev}_g : E_x \cong \mathrm{End}(V_{x,g})$, and for all g, g' the map*

$$\varphi : (g = g') \to \{\textit{bijections } V_{x,g} \to V_{x,g'} \textit{ intertwining } \mathrm{ev}_g, \mathrm{ev}_{g'}\}$$

(transport) is an equivalence, a loop going to its band scalar.

This is the result of the flattening, push-forward and normalisation steps (Gabber [2], Ch. II); we use it as given here, and compare its proof with the literature in Section 6. Its last clause says that $\Phi_x : G_x \to \mathrm{Split}(E_x)$, $g \mapsto (V_{x,g}, \mathrm{ev}_g)$, is fully faithful. We now upgrade it to an equivalence.

Proposition 4.2. *Let G be a banded groupoid, V and E as in Theorem 4.1, and $\Phi : G \to \mathrm{Split}(E)$, $\Phi(g) = (V_g, \mathrm{ev}_g)$.*

1. *The arrow of $\mathsf{ap}_\Phi(p)$ is $\varphi(p)$; hence Φ is an embedding.*
2. *Φ preserves the bands.*
3. *If G is a sheaf and $\Diamond G$, then Φ is an equivalence, and so is $j^{-1}\Phi : G \simeq \mathrm{Split}_0(E)$, which preserves the bands.*

Proof. (1) By path induction, both are the identity at refl. So ap_Φ is φ followed by the inverse of the equivalence of Proposition 3.2, an equivalence. (2) $\beta_{\Phi g}(\mathsf{ap}_\Phi \ell)$ is the scalar of $\varphi(\ell)$, which is $\beta_g(\ell)$. (3) The fibre $\sum_g (\Phi g = o) \in \mathcal{U}_1$ is a proposition by (1), and a sheaf (Σ of the sheaf G and the sheaves $\Phi g = o$, Theorem 3.6). For $g : G$ it holds, since Φg and o are merely

isomorphic (Skolem–Noether, as in Theorem 3.6); as $\Diamond G$, it holds (Proposition 1.1, eliminating into a sheaf proposition of $\mathcal{U}_1$). For the bands of $j^{-1}\Phi$: $\mathsf{ap}_j \mathsf{ap}_{j^{-1}\Phi} = \mathsf{ap}_{jj^{-1}\Phi}$, and $jj^{-1}\Phi = \Phi$; by induction on this path the bands are those of Φ, which are those of G by (2). □

Theorem 4.3 (H^2 form, object hypothesis)**.** *Let A be finitely presented and $G : \operatorname{Spec} A \to$ gerbes such that each G_x is a sheaf (so G is a family of stack gerbes), with an element of $\prod_{x:\operatorname{Spec} A} k_* G_x$ for some $k \geq 1$. Then merely there are N and Azumaya algebras $E : \operatorname{Spec} A \to \mathrm{AZ}_N$ with*

$$\delta(E) = \mathrm{cl}(G) \quad \text{in } H^2(\operatorname{Spec} A),$$

where $\delta(E) = \mathrm{cl}(x \mapsto \mathrm{Split}_0(E_x))$.

Proof. The goal is a proposition, so we may use the data of Theorem 4.1. Each $\mathrm{Split}_0(E_x)$ is a stack gerbe (Theorem 3.6). By Proposition 4.2, $j^{-1}\Phi_x : G_x \simeq \mathrm{Split}_0(E_x)$ is a band-preserving equivalence, so $G_x = \mathrm{Split}_0(E_x)$ by Proposition 2.2, and the classes agree. □

5. The group H^2 and torsion classes

Line families. A *charged type* is a type $X \in \mathcal{U}_0$ with $\Diamond X$, any two points merely equal, and a function w from loops of X to $R^\times$ (not required to be a band). $L(X, w)$ is the type of families $V : X \to \mathrm{Mod}$ whose fibres merely have a basis of rank one and on which each loop ℓ acts by multiplication by $w(\ell)$. A stack gerbe G is a charged type with $w = \beta$; the band change of Section 4 is $k_* G = L(G, \beta^k)$.

Proposition 5.1. *If $L(X, w)$ has an object over each point of X, then it is (equivalent to) a stack gerbe.*

Proof. Paths of $L(X, w)$ are families of linear bijections (the total space is contractible), so $L(X, w)$ is a 1-type. An automorphism of V is a scalar on each fibre; the scalars are equal along paths, and X is connected, so they glue (in the sheaf R, using $\Diamond X$) to one unit: this is the band. The isomorphisms $V \cong V'$ form an $R^\times$-torsor which is a sheaf; it is locally inhabited, since at a point x a bijection $V_x \cong V'_x$ extends to an invariant section of $\mathrm{Hom}(V, V')$ (weight 1, so it descends), hence (Hilbert 90) any two objects are merely equal. Descent for objects: a family f over a generator G_i (a proposition) glues to $x \mapsto \prod_{g:G_i} f(g)_x$, which has weight w (transport is pointwise) and fibres of rank one locally, hence merely (Theorem 1.3). Size: rank-one modules form a locally small type whose objects are merely R, so by replacement it has a small model Line_0, and families $X \to \mathrm{Line}_0$ with the weight condition give a small type mapping onto $L(X, w)$; by replacement again $L(X, w)$ has a small model, and the structure transfers along it. □

Proposition 5.2 (maps)**.** *A band-preserving map $e : G \to G'$ of stack gerbes is an equivalence, so $G = G'$.*

Proof. On loops $\beta' \circ \mathsf{ap}_e = \beta$ with β, β' equivalences. For a, b, being an equivalence $\mathsf{ap}_e : (a = b) \to (ea = eb)$ is a sheaf proposition, and $\Diamond(a = b)$, so by path induction it reduces to loops. The fibres of e are then propositions and sheaves, and locally inhabited, hence inhabited. Conclude by Proposition 2.2. □

Three maps produce all comparisons below. (1) *Pull-back*: if $\varphi : X \to Y$ satisfies $w_Y(\mathsf{ap}_\varphi \ell) = w_X(\ell)$, then $V \mapsto V \circ \varphi$ maps $L(Y, w_Y) \to L(X, w_X)$ and keeps the bands (transport along ℓ in $V \circ \varphi$ is transport along $\mathsf{ap}_\varphi \ell$ in V). (2) *Evaluation*: $x \mapsto (V \mapsto V_x)$ maps X to $L(L(X, w), \beta)$ and carries w to the band. (3) *Points*: if $k : K$, then $x \mapsto \mathrm{Line}(k = x)$ maps K to the trivial gerbe $T = L(1, 1)$ (rank-one modules) and keeps the bands.

Definition 5.3. $DG = L(G, \beta)$; $G+H = D\,L(G\times H, \beta\times\beta)$, where $(\beta\times\beta)(\ell) = \beta_G(\ell_1)\beta_H(\ell_2)$; $0\cdot G = T$, $(n+1)\cdot G = G + n\cdot G$.

(The twisted line $\mathrm{Line}(s = -)$ is an object of DG, and $s \mapsto \mathrm{Line}(s = -)$ inverts the band, so D reverses the class; this is why the dual appears in the sum.)

Theorem 5.4 (group law)**.** *For stack gerbes, $DDG = G$, $G + H = H + G$, $(G + H) + K = G + (H + K)$, $G + T = G$, $G + DG = T$, and $n \cdot G = D(n_*G)$. Hence $H^2(X)$, with $\mathrm{cl}(G) + \mathrm{cl}(H) = \mathrm{cl}(x \mapsto G_x + H_x)$, $-\mathrm{cl}(G) = \mathrm{cl}(DG)$ and $0 = \mathrm{cl}(x \mapsto T)$, is an abelian group, and*

$$n \cdot \mathrm{cl}(G) = \mathrm{cl}(x \mapsto D(n_*G_x)).$$

Proof. $DDG = G$ by evaluation, and a gerbe with a point is T by (3); DT has the point $V \mapsto V_*$, so $DT = T$. Commutativity: pull back along the swap. Unit: $g \mapsto (g, *)$ carries ℓ to (ℓ, refl), so $L(G\times T) = DG$ and $G+T = DDG = G$. Inverse: $(g, V) \mapsto V_g$ is a point of $L(G\times DG, \beta\times\beta)$, so it is T and $G+DG = DT = T$. Associativity: both $L((G+H)\times K)$ and $L(G\times(H+K))$ are pull-backs to $G\times H\times K$ (with weight $\beta\beta\beta$) along a product of an evaluation and an identity. Multiples: the constant family R is a point of 0_*G; and $g \mapsto (g, \mathrm{ev}\, g)$, $G \to G\times D(n_*G)$, carries ℓ to $\beta(\ell)\beta(\ell)^n$, so $L(G\times D(n_*G)) = (n+1)_*G$ and $(n+1)\cdot G = G + D(n_*G) = D((n+1)_*G)$. The operations respect mere equality, so they descend to the set quotient, and the laws, being propositions, are checked on classes. □

Proposition 5.5. $\mathrm{cl}(K) = 0$ *if and only if* $\|\prod_{x:X} K_x\|$.

Proof. The quotient is effective, so $\mathrm{cl}(K) = 0$ gives $\|K = (x \mapsto T)\|$, a mere path of maps, along which the point of T moves to a global section. Conversely a global section makes each K_x trivial by (3). □

The truncation surrounds the whole family: $\prod_x \|K_x\|$ would not suffice.

Theorem 5.6 (Gabber, H^2 form)**.** *Let A be finitely presented and $G : \mathrm{Spec}\, A \to$ gerbes with each G_x a sheaf. If $k \cdot \mathrm{cl}(G) = 0$ in $H^2(\mathrm{Spec}\, A)$ for some $k \geq 1$, then merely there are N and $E : \mathrm{Spec}\, A \to \mathrm{AZ}_N$ with $\delta(E) = \mathrm{cl}(G)$.*

Proof. By Theorem 5.4, $\mathrm{cl}(x \mapsto D(k_*G_x)) = 0$, so by Proposition 5.5 merely each $D(k_*G_x)$ has a point, uniformly in x. A point makes $D(k_*G_x)$ trivial, so $k_*G_x = DD(k_*G_x) = DT = T$ has a point. This is the hypothesis of Theorem 4.3, and the goal is a proposition. □

5.1. **Stack gerbes are the sheafification of $K(R^\times, 2)$.** Let K be the type of $R^\times$-gerbes in the plain sense: 1-types $X \in \mathcal{U}_0$ with $\|X\|$, $\|a = b\|$ for all a, b, and a band at every point. Let T_0 be the gerbe of $R^\times$-torsors (made small by replacement), with point k_0 the trivial torsor.

Proposition 5.7. *K is $K(R^\times, 2)$: any two points of K are merely equal, and $(T_0 = T_0) \simeq \mathrm{Tor}(R^\times)$, the type of $R^\times$-torsors.*

Proof. As in Proposition 2.2, a band-preserving equivalence $e : G \simeq G'$ gives $G = G'$, and if moreover $e(g) = g'$ then $(G, g) = (G', g')$ as pointed gerbes. If $g : G$, then $y \mapsto (g = y)$ is a band-preserving equivalence $G \simeq T_0$ sending g to k_0. Hence any G is merely equal to T_0 (the goal is a proposition, so we may take a point), and $\sum_{G:K} G$ is contractible with centre (T_0, k_0). By the fundamental theorem of identity types, transport of k_0 is an equivalence $(T_0 = G) \simeq G$; at $G = T_0$ this is $(T_0 = T_0) \simeq T_0 \simeq \mathrm{Tor}(R^\times)$. □

Proposition 5.8. StackGerbe *is a sheaf.*

Proof. It is a retract of $\sum_Y D(Y)$, where Y ranges over the sheaves in $\mathcal{U}_0$ and $D(Y)$ is the band, the 1-type condition, $\Diamond Y$ and $\Diamond(a = b)$: the paths of a sheaf are sheaves, and the remaining data are propositions. The sheaves of $\mathcal{U}_0$ form a sheaf (Proposition 1.1). Each $D(Y)$ is a sheaf, being built by Π, Σ and identity types from Y, its paths, $R^\times$ (a sheaf, as R is) and the sheaf propositions $\Diamond(-)$. Sheaves are closed under Σ and retracts. □

The closure works on small propositions; for a type F in $\mathcal{U}_1$ we say that F holds *locally* if there is a small A with $\Diamond A$ and a map $A \to F$. A map f between 2-types is $\bigcirc$*-connected* if each fibre F of f holds locally, and so do $x = y$, $p = q$ and $\alpha = \beta$ for all $x, y : F$, $p, q : x = y$ and $\alpha, \beta : p = q$. This is $\bigcirc$-connectedness of the fibres for the modality $\bigcirc$ of the root topology [8]: $\bigcirc$ is topological, hence lex, so it preserves identity types and $\bigcirc F$ is contractible iff $\Diamond F$ and each $\bigcirc(x = y)$ is contractible; the paths at depth 3 are propositions, for which $\bigcirc$-connected means holding locally.

Theorem 5.9. *The map $\eta : K \to$ StackGerbe that weakens $\| - \|$ to $\Diamond$ is defined, and it is $\bigcirc$-connected. Hence* StackGerbe *is a sheaf with a $\bigcirc$-connected map from $K(R^\times, 2)$, that is,* StackGerbe $= \bigcirc K(R^\times, 2)$*, and $H^2(X) = \|X \to \bigcirc K(R^\times, 2)\|_0$, with $\bigcirc$ applied to K, not to the set truncation. Post-composition with η induces $\|X \to K\|_0 \to H^2(X)$.*

Proof. η is defined: a gerbe G with a point g maps to the trivial stack gerbe T of lines by $x \mapsto \mathrm{Line}(g = x)$, keeping the bands (map (3)); T is a gerbe, so the map is an equivalence, and the carrier of G is a sheaf (being a sheaf is a proposition, so a point may be assumed), hence also its paths. Next, $K \simeq \sum_{S:\mathrm{StackGerbe}} \|S\|$: send G to $(\eta G, \|G\|)$, and (S, t) to the gerbe with carrier S, inhabited by t, connected by Hilbert 90. Both round trips change only propositions. So the fibre of η at S is that of the first projection, the proposition $\|S\|$. It holds locally, since $\Diamond S$ and a point of S gives a point of the fibre; and a proposition that holds locally is $\bigcirc$-connected. □

So η is an embedding: the gerbes are exactly the stack gerbes with a mere point, and η adds the stack gerbes that have points only locally.

6. Relation to the literature

The strategy is classical. For affine schemes, Br = Br′ is due to Gabber [2]. Hoobler [5] introduced, for regular rings, the local-to-global argument showing that the set of f over which the class is represented is an ideal; Hoobler [3] simplified Gabber's proof, and Lieblich [10, §3.1.4] simplified it further with twisted sheaves. De Jong [7], for schemes with an ample line bundle, takes the affine case as input (his Lemma 3.4 quotes Hoobler's Theorem 7). The skeleton of Theorem 4.1 is Lieblich's: twisted locally free sheaves on the gerbe, End at the end; the ideal argument with gluing over $D(f) \cup D(g)$; reduction of the étale-local to the Zariski-local situation by Gabber's splitting-algebra lemma and push-forward along finite locally free covers ([10, 3.1.3.5–6]; our flattening by towers of root adjunctions plays this role); K-theory through the nilpotence of the rank-zero part of K_0 and roots in $K_0 \otimes \mathbb{Q}$. The identification $\delta(E) = \mathrm{cl}(G)$ of Section 4 is the standard dictionary between gerbes and splittings; here it is formulated and checked internally.

The gluing step is different. The arguments of Gabber, Knus–Ojanguren and Lieblich use, in the gluing, the following facts about $K_0(A)$:

- *effectiveness*: some multiple of a class of positive rank is the class of a projective module;
- *cancellation after taking direct sums*: for finite projective modules P, Q of positive rank, $[P] = [Q]$ implies $P^{\oplus m} \cong Q^{\oplus m}$ for some $m \geq 1$.

Equality of classes already gives a stable isomorphism $P \oplus A^r \cong Q \oplus A^r$ by the definition of K_0; this elementary fact is distinct from the cancellation result used to remove the added summand after taking multiples. Specifically:

- Gabber [2], Ch. II, Lemma K, proves both by reducing to finitely generated $\mathbb{Z}$-algebras and applying "Serre's theorem and the cancellation theorem" (Bass); the gluing argument that follows invokes Lemma K(ii) and (iii).
- Knus–Ojanguren [9], Lemma 3.2, realise rational K_0-classes by projective modules and then apply "le théorème d'effacement de Bass" (Bass, Chap. IX, §7).
- Hoobler [5] (regular case) realises classes by modules of large rank using Bass, Chap. IX, 4.1.
- Lieblich [10, 3.1.4.3, 3.1.4.4] uses Bass cancellation and Serre splitting, after the same reduction, to realise n-th roots in $K_0 \otimes \mathbb{Q}$ by bundles before gluing.

Our gluing (companion note `gabber-recollement.tex`) avoids general effectiveness and Bass cancellation. It consists of two parts.

(1) *Weak cancellation, proved elementarily.* For twisted bundles of positive rank, $X \oplus T \cong Y \oplus T$ implies $X^{\oplus 2m} \cong Y^{\oplus 2m}$ for some $m \geq 1$: once T is a summand of $X^{\oplus m}$ and of $Y^{\oplus m}$ (by evaluation and projectivity), $X^{\oplus(m+1)} \cong X^{\oplus m} \oplus Y$, and substitution gives $X^{\oplus 2m} \cong X^{\oplus m} \oplus Y^{\oplus m} \cong Y^{\oplus 2m}$. The conclusion is of a known type (it is the form of cancellation in Lieblich's Lemma 3.1.4.3(ii)); the difference here is the elementary proof, which needs no dimension estimate or stable range.

(2) *Goodness as the induction invariant, and an explicit clutching correction.* One keeps $\rho(P(V)) = 1$ in $K_0 \otimes \mathbb{Q}$ for $P(V) = V^{\otimes n} \otimes \Lambda^{-1}$, $\rho(F) = [F]/\operatorname{rk} F$. After gluing two good bundles, the defect $\tau = \rho(P(V')) - 1$ lies in the kernel of restriction to both pieces, hence $\tau^2 = 0$ by a clutching computation (Whitehead's identity and Quillen patching, Lombardi–Quitté XVI.3.4). An actual bundle S with $\rho(S) = 1 - \tau/n$ is obtained from an inverse power of the transition matrix plus a free summand, and $\rho(P(V' \otimes S)) = (1+\tau)(1-\tau/n)^n = 1$. No arbitrary positive-rank rational class ever has to be realised by a bundle; the only class that must be realised has a directly constructed representative.

The second part is the substantive simplification: it removes effectiveness, Bass cancellation, Serre splitting and the reduction to noetherian rings from the gluing argument. This construction differs from the passages examined in Gabber, Knus–Ojanguren, Hoobler (1980) and Lieblich. This comparison establishes a simplification of those arguments, rather than historical priority for the construction. The constructive characterisations of Azumaya algebras in [6] concern a different result.

The result is a constructive, machine-checked internal proof of the gerbe form stated in Theorem 4.1, following the classical strategy. General effectiveness and Bass cancellation are replaced by elementary weak cancellation and an explicit clutching correction. The H^2 statement of Theorem 4.3 follows by the identification of Sections 3–4.

7. What is formalised in Unit

Everything above is checked in Unit, in `unit/sag/twisted`: `./segc sag/twisted all` gives 418 modules, 0 failed, about 12 s. The trees that link the changed files also check: `sag/torsion-c` (418 modules), `sag/gabber-end` (383) and `sag/split-b` (389). No new axiom is added; the Theory parameters are univalence, function extensionality and propositional univalence (both derived from it) and replacement (images and their surjectivity; they give

$\|-\|$, set quotients and Split_0), together with the SAG axioms at one universe level and Zariski locality. The closure $\Diamond$ is constructed (Proposition 1.1); it is not a parameter.

Statement	File
Lemma 1.2 and bases $\leftrightarrow$ frames	`BasisFrames`
Theorem 1.3	`DescentTheoremD` (earlier)
Proposition 1.1	`EtaleClosure`, `TowerDescent`, `Flattening`
Proposition 2.2, H^2, cl	`StackGerbes` (`Eq`, `H2`)
Proposition 3.2	`SplitSIP`
Proposition 3.3	`SplitSmall`
Lemma 3.4	`EndCentre`
Proposition 3.5	`SplitBand`
Theorem 3.6, local connectedness	`SplitConnected`
Theorem 3.6, the rest	`SplitGerbe`
Proposition 5.1	`BandChange`, `BandChangeGerbe`, `LineGerbe`
Proposition 5.2 and small models	`GerbeMaps`
maps (1)–(3), D, T	`GerbeOps`
$G + H$	`GerbeSum`
Theorem 5.4, pointwise laws	`GerbeLaws`
Theorem 5.4 on H^2, Proposition 5.5	`H2Group`
Theorem 4.1	`GabberAzumaya` (earlier)
Proposition 4.2	`SplitEquivalence`
Theorem 4.3	`GabberH2`
Theorem 5.6	`GabberTorsion`
Proposition 5.8	`StackSheaf`
Proposition 5.7	`GerbeSpace`
Theorem 5.9	`StackComparison`

The final declaration is

`GabberTorsion.Closed[R,setR,axioms,loc].At[A,fp,G,stk,n0,torsion].theorem`

of type $\|\sum_n \sum_{E:\mathrm{Spec}\,A\to \mathrm{AZ}_{n+2}} \delta(E) = \mathrm{cl}(G)\|$, where `stk` says that each G_x is a sheaf and `torsion` is an equality $(n_0+1)\cdot \mathrm{cl}(G) = 0$ in the group $H^2(\mathrm{Spec}\,A)$ (`H2Group`); it instantiates `GabberTorsion.Over`, stated for any closure at level 0, with the constructed one. Theorem 4.3 is `GabberH2.Closed`. Negative tests are kept: composing scalar arrows in the wrong order, Φ sending a loop to the inverse of its band, eliminating $\Diamond P$ into $\|P\|$ without a sheaf hypothesis, evaluation carrying w to the inverse of the band, taking the weight w_W/w_V of $\mathrm{Hom}(V,W)$ to be $\beta\times\beta$ by computation (it is so only up to a proof), and taking the second component of $\mathsf{ap}_{g\mapsto(g,*)}\ell$ to be refl by computation, taking a loop of K at T to fix k_0 by computation, and reading a stack gerbe as a gerbe ($\Diamond S$ in place of $\|S\|$), are all rejected. Theorem 5.9 is `StackComparison.Over[R,setR,axioms,loc,sh]: localisation` (sheaf and $\bigcirc$-connected, with `eta-connected`) and `Classes[X].map`; Proposition 5.7 is `GerbeSpace.Over` (`connected`, `loops`).

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